Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 7 Extension

Total _____ */ 23*

A. Which equation? $a = \frac{v-u}{t}$ *or* $v^2 - u^2 = 2\,a\,s$ *— the empty line in your Givens box tells you.*

1) **Calculate** the speed a go-kart was moving at before it accelerated at $2.5\,\text{m/s}^2$ for $4.0\,\text{s}$, reaching $18\,\text{m/s}$. **[3]**

Givens

$s =$
$u =$
$v =$
$a =$
$t =$

2) **Calculate** the acceleration of a motorbike that brakes from $24\,\text{m/s}$ to a stop in $48\,\text{m}$. **[3]**

Givens

$s =$
$u =$
$v =$
$a =$
$t =$

3) **Calculate** the speed of a hockey puck after it slides $40\,\text{m}$ across the ice. It is hit at $5.0\,\text{m/s}$ and slows down with an acceleration of $-0.20\,\text{m/s}^2$. **[3]**

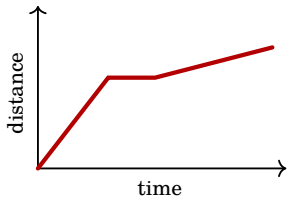
Givens

$s =$
$u =$
$v =$
$a =$
$t =$

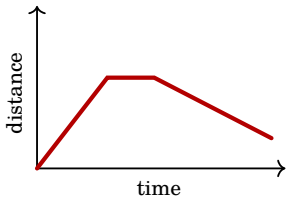
B. Graphs *choose the graph that matches.*

4) A bus drives at a steady speed, stops at a bus stop, then drives on **faster**. Which distance–time graph shows this journey? **[2]**

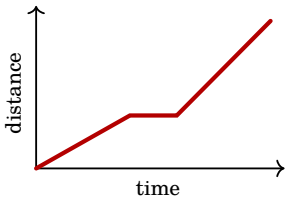
[1]



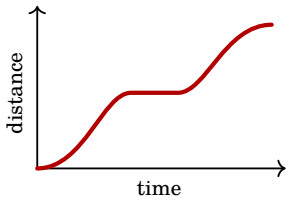
[2]



[3]



[4]

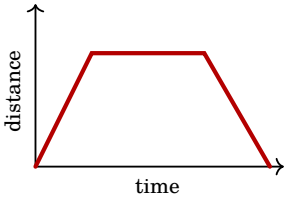


Graph _____ because

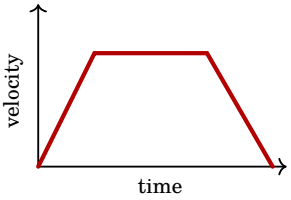
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5) A lift speeds up steadily from rest, moves at a constant speed, then slows steadily to a stop. Which graph shows this? **[1]**

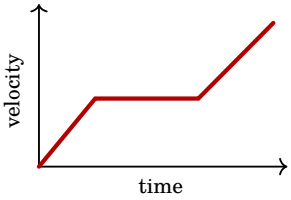
[1]



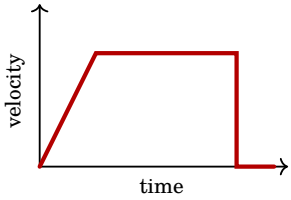
[2]



[3]

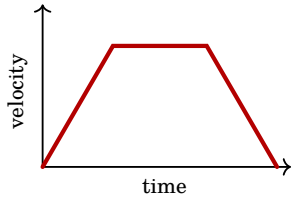


[4]

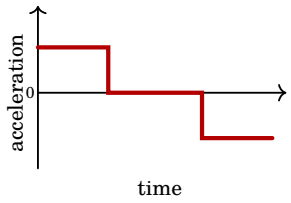


Graph _____

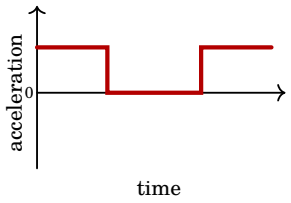
6) Here is a velocity–time graph. Which acceleration–time graph matches it? **[1]**



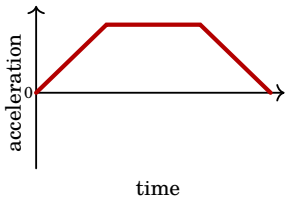
[1]



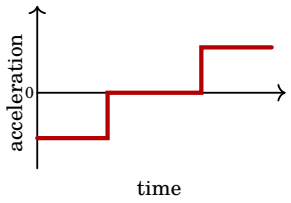
[2]



[3]

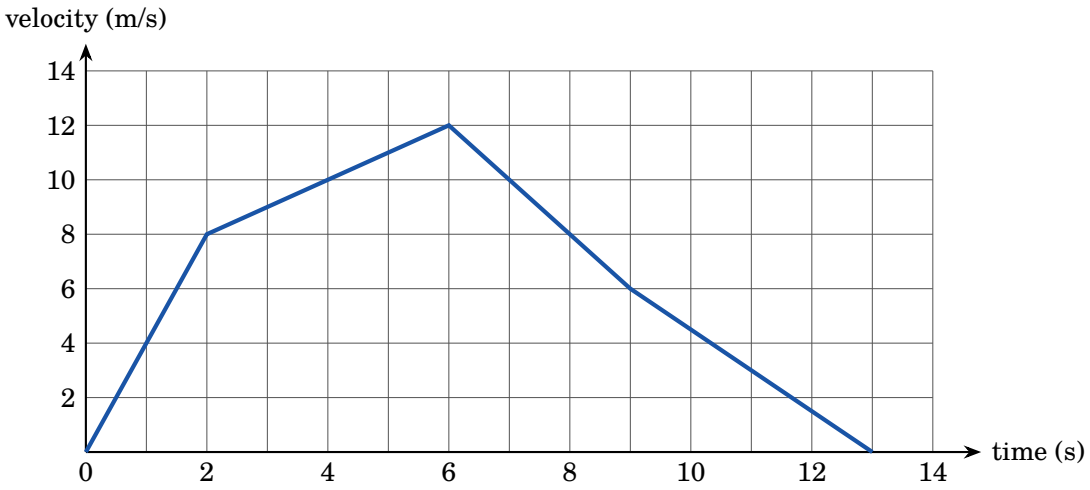


[4]



Graph _____

7) A motor boat leaves a jetty, then slows down to moor. The graph shows its motion.



(a) Between which times is the acceleration greatest? Explain how the graph shows this.

[2]

(b) Calculate the acceleration between 2 s and 6 s.

[1]

(c) Describe the motion of the boat from 6 s to 13 s.

[2]

(d) Use $v^2 - u^2 = 2\,a\,s$ and your answer to (b) to calculate the distance travelled between 2 s and 6 s.

Givens

$s =$
$u =$
$v =$
$a =$
$t =$

[2]

(e) Use the area under the graph to calculate the total distance the boat travels. The area between 2 s and 6 s matches your answer to (d).

[3]